

# The Compiler Closure of TFPT

What concretely simplifies, reduces and closes relative to the seven original papers — a status assessment and reading guide

June 11, 2026

## In one sentence

The theory tips from “many sectors with surprising hits” (the state of Papers 1–7) to “*one small discrete compiler generates those sectors*”:  $E_8$  — which in v4.5 appeared only *peripherally* (the orbit cascade for scale ordering) — is now *built* from two TFPT atoms and moved to the centre; the flavor matrix, the fine-structure constant and the scale grammar follow as *fixed points* from exactly two inputs  $c_3 = \frac{1}{8\pi}$  and  $g_{\text{car}} = 5$ .

## The TFPT document set — what is treated where

**Plain language:** TFPT is a small discrete compiler. Two inputs — the seam constant  $c_3 = \frac{1}{8\pi}$  and the carrier rank  $g_{\text{car}} = 5$  — build  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  and read off the Standard Model, the constants and the scale grammar. The development is **five short documents plus Appendix H**, best read in order (this is the entry point; a detailed map is in §1):

1. **introduction** — reading guide, compiler closure, paper-by-paper comparison, predictions, the dependency DAG and proof ledger.
2. **tfpt\_1\_architecture\_e8** — the two axioms, the derivation map, the EM fixed point  $\alpha^{-1}$ , the  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  construction.
3. **tfpt\_2\_standard\_model** — the SM in one  $\varphi_0$ -ladder formula, flavor from parabolic transport, the worked closures, and gravity/QG as the seam response.
4. **tfpt\_3\_e8\_audit\_bootstrap** — the seven  $E_8$  slices as an audit raster, the cascade bridge, the Möbius bootstrap.
5. **tfpt\_4\_frontier** — honest status of  $\eta_B$ ,  $m_p/m_e$ , Koide, dark matter, full QG.
- H. **tfpt\_horizon\_readouts** — **Appendix H** (unit-system reframe, *not* new physics): one seam constant as the universal horizon thermal code.
- O. **origin\_theory** — **Origin Theory**: why no free fundamental number remains. The  $(g_{\text{car}}, N_{\text{fam}})=(5, 3)$  skeleton, the triply-forced 8 (geometry = lattice = gravity), the order-30 Coxeter cycle, one boundary transport for flavor and horizon, and the gapped *unique attractor*. [I]/[L] core (v53–v56) plus one honestly-typed [P] cyclic interpretation.

You are reading document #1.

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**Reading the markers (refined).** The former overloaded “[M]” is split into the grade it actually carries, so a reviewer can see at a glance which claims are exact, which are formal, and which are conditional: **[I]** *exact identity* (e.g.  $240 = 16 \cdot 5 \cdot 3$ ); **[L]** *Lie/lattice theorem* (e.g.  $D_5 \oplus A_3 + \mu_4 = E_8$ ); **[F]** *formalised* (Lean- or script-verified); **[N]** *numerical fixed point* (reproduced, stable); **[P]** *physical/conditional* (follows under named hypotheses); **[A]** *axiom or open*.

|                                    |                                              |
|------------------------------------|----------------------------------------------|
| Physical — under hypotheses        | RG masses, QG    QG.AMB.01                   |
| Numerical — fit-free fixed point   | $\alpha^{-1}, \Lambda, \theta_{12}$ EM.FP.01 |
| Lattice — Lie / glue / root system | $E_8 = (D_5 \oplus A_3) + \mu_4$ E8.GLU.01   |
| Formal — Lean / exact script       | P2 algebra    FORM.P2.02                     |
| Axiom — declared inputs            | P1 $c_3$ , P2 $g_{\text{car}}$ AX.P1.01      |

The *claim stack* (bottom = most certain/fewest, top = conditional). Each claim sits on exactly one level; the bottom two are inputs, the middle two are proved/reproduced, only the top is conditional. The single source of truth for which claim sits where is **verification/status\_ledger.csv** (Fig. 1).

**No-free-pattern rule.** To keep the structure from drowning in decorative coincidences, an identity enters the *main text* only if it (1) is a necessary step in a derivation, (2) kills an alternative, (3) is ablation-relevant, (4) links two previously independent modules, or (5) is experimentally tested. Everything else is explicitly demoted to an *audit fingerprint* **[I]** (clearly boxed as such) — so the load-bearing beams stay visible and the ornament never gets promoted to spine.

**Admissible-invariant classes (the harder filter).** Sharper still: a *load-bearing* integer may only come from one of seven typed sources — (i) a symmetric function  $e_k(a)$  of the anchor  $a = (1, 1, 2)$ ; (ii) a power sum  $p_n(a)$  or difference  $p_{n+1} - p_n$ ; (iii) a determinant, spectrum, Smith normal form or trace of  $(R, Q, K, L)$ ; (iv) an anchor quadratic form  $u^\top M v$  with  $u, v \in \{1, a\}$ ; (v) a Plücker coordinate of the anchor plane  $\langle 1, a \rangle$ ; (vi) an  $E_8$  Casimir degree or branching block; (vii) a pre-registered experimental observable. Anything else is an audit fingerprint or a frontier item, never spine. This is what converts TFPT from *pattern hunting* (“look, another number”) into a typed compiler: *fewer admissible kinds of explanation*, not more matches.

#### Reviewer path (read in this order; the rest is support)

1. the architecture: the two axioms  $\{c_3, g_{\text{car}}\}$  and the two-engine picture (§1, the figures);
2. the  $E_8$  glue  $E_8 = (D_5 \oplus A_3) + \mu_4$  — document 1, Part II, machine-checked

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(verification/v1_e8_glue.py);
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3. the  $\alpha^{-1}$  fixed point plus ablation (v3\_em\_alpha.py);
4. the **single status ledger** verification/status\_ledger.csv — the source of truth for every claim’s grade, location, dependency and script;
5. the honest frontier list (document 4).

Everything carries a claim ID (e.g. E8.GLU.01, EM.FP.01) that resolves to a row of the ledger. **If the text and the ledger ever disagree, the ledger wins.**

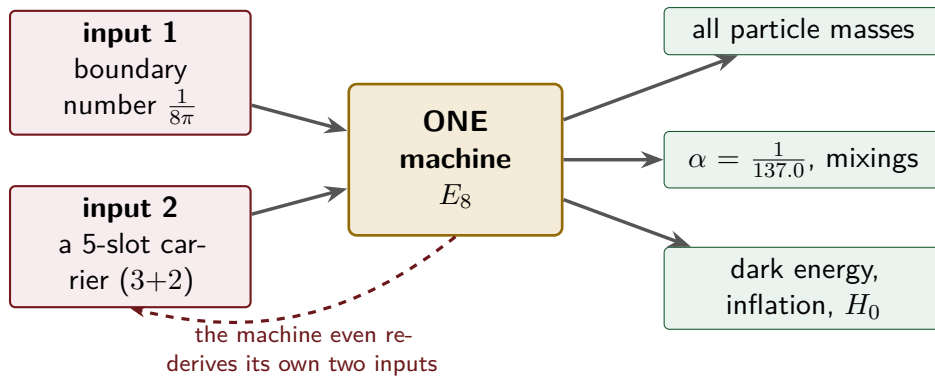
## 1 What this is about: the jump in construction principle

### In plain words — what we found, and why it matters

**What we found.** The older version of the theory already reproduced dozens of real numbers of physics (particle masses, the fine-structure constant, dark-energy and inflation values) from a small set of inputs. What it could *not* say was *why* the same handful of small integers (2, 3, 5, 16, ...) kept reappearing in every sector. The new result closes exactly that gap: those numbers are not separate lucky hits. A single tiny “machine” — fed by just **two inputs** (a boundary number  $\frac{1}{8\pi}$  and a five-slot carrier) — builds the exceptional symmetry  $E_8$ , and  $E_8$  is the unimodular “audit hull” that classifies the discrete Standard-Model readout: the constants, the flavor skeleton and the gravity/cosmology regime are then read off by projection and boundary dressing (not “printed” for free — where a scheme or analytic interface is needed, it is named).

**How this drastically simplifies the theory.** Instead of seven separate derivations that happen to share numbers, there is now *one generator and a handful of corollaries*. Roughly 250 pages of sector-by-sector work collapse to “two inputs + one machine”; the number of independent structural assumptions drops to **two**.

**How plausible it is.** A theory becomes more believable when it *explains more from less*. Here the two inputs are even *over-determined*: the machine reproduces the very two numbers it started from (the dashed loop below), so there is *no free dial left to tune*. The only genuinely free, continuous ingredient that remains is the number  $\pi$ .

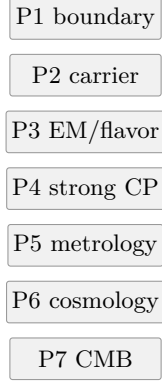


*Two numbers go in on the left; the discrete Standard-Model core, the dimensionless constants and several scale readouts come out on the right — and the dashed loop is the point: the machine reproduces the very two numbers it started from, so the discrete core is overdetermined rather than fitted. (Dimensionful EW/QCD masses, the full quantum-gravity measure and the frontier items stay typed-open — see the status ledger.)*

The seven original papers (tfpt-45, Papers 1–7) deliver a long chain of physical closures: boundary kernel → carrier → Standard-Model packet → EM fixed point → flavor transport →

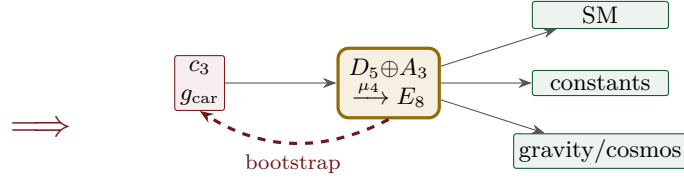
admissibility/strong CP  $\rightarrow$  geometric gravity/metrology  $\rightarrow$  cosmology/CMB. Each stage stands on its own and hits a remarkable number of values. What was missing was a *meta-statement* explaining *why* so many independent sectors use the same small stock of numbers (2, 3, 5, 16, 40, 41, 48, 240, 248).

#### Version 4.5 — seven papers



many parallel  
sector closures;  
 $E_8$  only a **periph-**  
**eral** backdrop

#### New — one compiler



two inputs build  $E_8$ ;  
 $E_8$  feeds back and fixes  
the inputs (closed loop)

**The difference in one picture.** v4.5 is a *stack of seven independent closures* that happen to share numbers, with  $E_8$  assumed in the background. The new approach is *one discrete compiler*: two inputs build  $E_8$ ,  $E_8$  outputs the sectors, *and*  $E_8$  feeds back to fix the two inputs (the red loop). What were seven parallel results become one generator with corollaries; what was a posited  $E_8$  becomes a constructed-and-self-checking closure.

#### Sharper still: one anchor $a = (1, 1, 2)$ , and $E_8$ as a *compiler hull* [1]/

The two axioms are not even independent: they are the *elementary symmetric polynomials* of the single parabolic anchor  $a = (1, 1, 2)$  (the exponents-at-infinity / splitting type  $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ ):

$$e_1(a) = 4 = |\mu_4|, \quad e_2(a) = 5 = g_{\text{car}}, \quad e_3(a) = 2 = |\mathbb{Z}_2|, \quad c_3 = \frac{1}{2e_1(a)\pi} = \frac{1}{8\pi}.$$

Its *power sums*  $p_n(a) = 1^n + 1^n + 2^n = 2 + 2^n$  generate the big Lie data directly:  $p_1=4=|\mu_4|$ ,  $p_2=6=|R^+(A_3)|$ ,  $p_3=10=\mathcal{A}_\Lambda$ , and

$$|R(E_8)| = p_1 p_2 p_3 = 240, \quad \dim E_8 = p_1 p_2 p_3 + (p_4 - p_3) = 248, \quad p_4 - p_3 = 8 = \text{rank } E_8,$$

with the binary ladder  $p_{n+1} - p_n = 2^n$ . So the inputs collapse from  $\{c_3, g_{\text{car}}\}$  to  $a = (1, 1, 2)$  plus  $\pi$ . [verification/v23\_anchor\_generator.py]

And  $a$  is itself *half-forced* ([L]/[P]): on  $\mathbb{P}^1$  every bundle splits (Birkhoff–Grothendieck), so once  $|\mu_4| = 4$ ,  $N_{\text{fam}} = 3$ ,  $\deg E = -4$  are set, the three positive anti-degrees are positive integers summing to 4, and  $4 = 2 + 1 + 1$  is the *unique* partition into three positive parts. The genuine remaining axiom is therefore not “why  $a = (1, 1, 2)$ ” but “why the carrier  $g_{\text{car}} = 5 / (|\mu_4|, N_{\text{fam}}) = (4, 3)$ ” — that interface (with  $c_3$ ) is the [A] input layer.

**What  $E_8$  is (and is not).** In TFPT  $E_8$  is *not* an unbroken physical gauge group: it is the unimodular *audit* / *compiler hull* that classifies the admissible discrete charge and residue structures. The Standard Model is a *readout* after projection, not a direct  $E_8$  gauge theory; the Lorentz signature appears only after projection; fermions are not “everything in the adjoint”. This is exactly why the known no-go results against literal  $E_8$  world-formulas (Distler–Garibaldi; Coleman–Mandula) do *not* bite: they constrain  $E_8$  as a spacetime gauge symmetry, which

TFPT does not claim. The defensible claim is: *TFPT is the unique parabolic compilation of the anchor  $a = (1, 1, 2)$  into the  $E_8$  audit hull.*

In one line:

The anchor generates the syntax; the boundary generates the scale;  $E_8$  checks the consistency.

i.e. the discrete content (carrier  $D_5 \oplus A_3 + \mu_4$ , the operators  $R, Q, K, L$ ) flows from  $a = (1, 1, 2)$ ; the continuous content ( $c_3 = \frac{1}{8\pi}$ ,  $\varphi_0$ ,  $\alpha^{-1}$ , the exponential scales) flows from  $\pi$ ; and  $E_8$  is the unimodular *checksum container*, not the world group.

**The anchor ratio triad** ([[verification/v119\\_review\\_validation\\_2.py](#)]). The three elementary symmetric values, normalised by the family count, are the theory's three recurring fractions [I]:

$$\frac{e_3}{p_0} = \frac{2}{3} \text{ (Koide branch / sheet ratio), } \quad \frac{e_1}{p_0} = \frac{4}{3} \text{ (seed gain), } \quad \frac{e_2}{p_0} = \frac{5}{3} \text{ (carrier branch)}$$

— the canonical flow's critical points are exactly  $(2, 5) = (e_3, e_2)$  with inflection  $\frac{7}{2} = \frac{e_3 + e_2}{2}$ : the fractions  $\frac{2}{3}, \frac{4}{3}, \frac{5}{3}$  are not scattered numbers but the three normalised anchor coefficients. Further micro-identities [I]:  $\Omega_{\text{adm}} = 2p_1p_2 = 48$ ,  $10b_1 = 1 + p_1p_3 = 41$ ,  $\Delta_Y = e_2^2 = p_0^2 + 2 \text{rank } E_8 = 25$ ,  $h(E_8) = e_3p_0e_2 = 30$ ,  $\text{rank } E_8 = p_0 + e_2$ .

That is now the case. The whole development is consolidated into **five companion documents** (plus this introduction), organised in layers that build on one another; this introduction is the entry point and reading guide. Each document collects the earlier notes of its layer into self-contained parts:

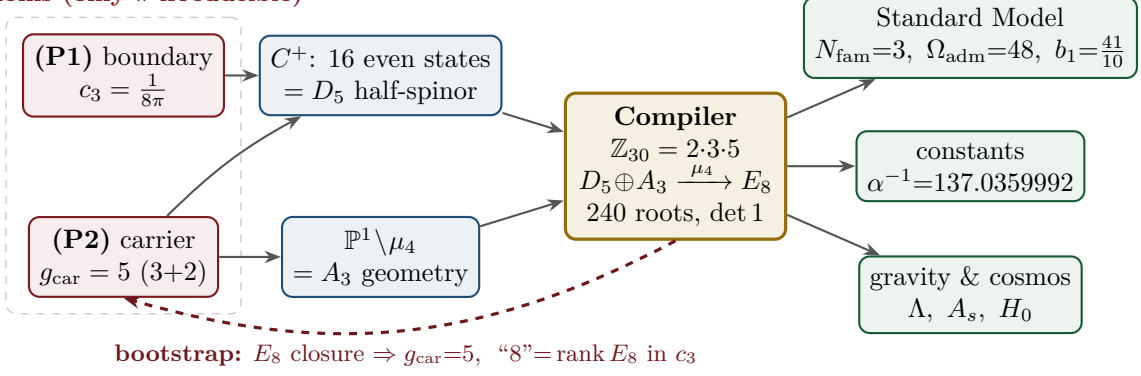
| Document                          | What it contains (former notes now as parts)                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| tfpt_1_architecture_e8            | <b>Core.</b> The two axioms $\{c_3, g_{\text{car}}\}$ , the derivation map and the EM fixed point (Part I); the Coxeter–cyclotomic compiler $\mathbb{Z}_{30}=2\cdot3\cdot5$ , the explicit $D_5 \oplus A_3 + \mu_4$ $E_8$ construction, the machine-checkable $E_8$ appendix and the group-level glue $(\text{Spin}(10) \times \text{SU}(4)) / \Delta \mathbb{Z}_4$ (Part II).           |
| tfpt_2_standard_model             | <b>Standard Model.</b> The full SM in one $\varphi_0^{\text{ret}}$ -ladder formula (Part I); the flavor block from parabolic weights and the transport theorem, H2 as a spectral selector (Part II); the worked closures — explicit gap, APS–Fredholm, truncation/ $n_s, r$ , Starobinsky $M$ , derived $\theta_{12}$ , quark $c$ , the H2 splitting type, $\Lambda$ -typing (Part III). |
| tfpt_3_e8_audit_bootstrap         | <b><math>E_8</math> audit &amp; bootstrap.</b> The seven $E_8$ slices as an audit raster ( $E_6 \times A_2$ reads the flavor matrix) (Part I); the old cascade $D=60-2n$ as the same even-integer spine (Part II); the Möbius bootstrap — $g_{\text{car}}=5$ and the “8” in $c_3$ as overdetermined fixed points, only $\pi$ irreducible (Part III).                                     |
| tfpt_4_frontier                   | <b>Frontier.</b> The honest status of $\eta_B$ , $m_p/m_e$ , Koide, dark matter and quantum gravity — the genuine handle, the precision, and what is <i>not</i> forced onto the ladder.                                                                                                                                                                                                  |
| tfpt_horizon_readouts<br>(App. H) | <b>Appendix H — horizon unit system (reframe, not new physics).</b> One seam constant $c_3 = \frac{1}{8\pi}$ as the universal horizon thermal code: Hawking/de Sitter/Unruh, BH thermodynamics, Page, scrambling, Nariai bound, $v_{\text{GW}}=c$ — standard physics in seam units, two compiler fingerprints ( $1920= W(D_5) $ , $ \mu_4 =4$ ).                                         |

The five formerly “research-level” problems are now *closed, reduced or typed* inside the documents where they belong — the group-level closure  $(\text{Spin}(10) \times \text{SU}(4)) / \Delta \mathbb{Z}_4$  in document 1 (closed), and the H2 splitting type, the Starobinsky scalaron mass, the  $\theta_{12}$  texture and the quark  $c$  in document 2 (the first two closed; the quark *ratios* now closed combinatorially (v49/v71), only the *absolute* scale a  $U_{\text{point}}$  anchor;  $\theta_{12}$  the seam-misalignment residual). Five companion files (plus this introduction) instead of a dozen, each result where it logically lives.

## 2 The architecture at a glance

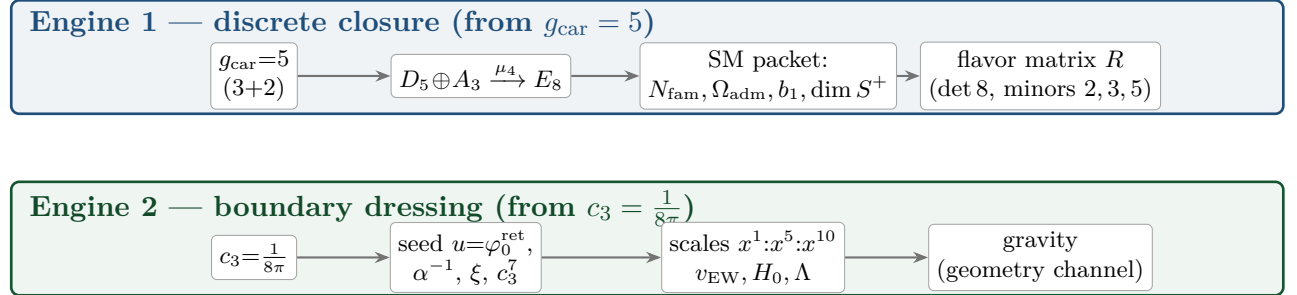
The diagram below shows the whole pipeline: *two* inputs on the left, the discrete compiler in the middle, the physics on the right. Everything in between is a consequence.

2 axioms (only  $\pi$  irreducible)



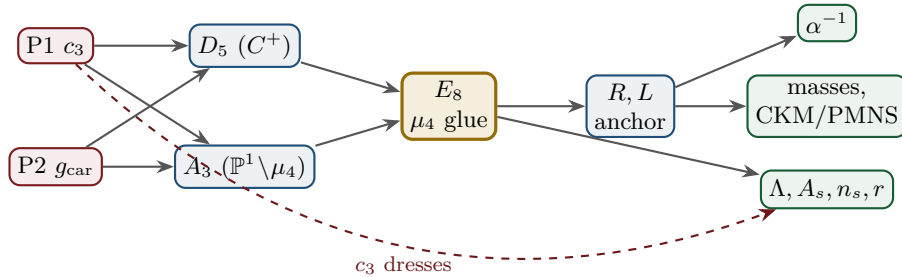
**Key statement (the loop closes):** previously  $D_5$ ,  $A_3$  and  $E_8$  sat *as given* Lie structures in the background. Now they are *intermediate products* of the two inputs — and the red back-channel is the new part: the  $E_8$  closure feeds back and *fixes the inputs*.  $g_{\text{car}} = 5$  is the unique rank-fill/Coxeter/Pascal solution ( $2^{g-1} = \binom{g}{0} + \binom{g}{1} + \binom{g}{2}$ ), and the integer 8 in  $c_3 = \frac{1}{8\pi}$  is rank  $E_8 = h(D_5) = \det R$ . **Inputs and output are mutually locked — they prove each other;** the only thing *not* produced by the loop is the continuous  $\pi$  (Möbius/Gauss–Bonnet), the lone irreducible primitive. So it is not a one-way arrow chain but a closed self-consistency loop.

**The master story is two engines.** Read from the two axioms, the whole theory factorises into exactly *two* engines — a *discrete* closure and a *boundary* dressing. Gravity is not a third block; it is the geometry channel of Engine 2.



### 3 The dependency DAG and the proof ledger

The whole theory is a directed acyclic graph: two axioms at the source, the compiler in the middle, the observables as sinks. The DAG makes the attack surface explicit — each arrow is a named theorem or a declared input, and each node fails in exactly one way.





| Node           | Definition                                             | Inputs      | St.     | Output                           | Failure mode                |
|----------------|--------------------------------------------------------|-------------|---------|----------------------------------|-----------------------------|
| P1             | RP boundary kernel, $c_3=1/(8\pi)$                     | — (axiom)   | [A]     | $c_3$                            | no reflection-positive seam |
| P2             | 5-slot carrier $g_{\text{car}}=5$                      | — (axiom)   | [A]     | carrier 3+2                      | wrong family/charge lattice |
| $D_5$          | $C^+$ even-Hamming = half-spinor 16                    | P2          | [I]     | $\dim S^+=16, Y$                 | group/matter mismatch       |
| $A_3$          | $\mathbb{P}^1 \setminus \mu_4$ four-puncture geom.     | P1          | [I]     | $N_{\text{fam}}=3, \mathbb{Z}_6$ | wrong multiplicity          |
| $E_8$          | $\mu_4$ glue: disc $=\mathbb{Z}_4$ , $q$ -sum=2        | $D_5, A_3$  | [I]     | 240, 248                         | not even-unimodular         |
| $R, L$         | residue + winding matrix, anchor (1, 1, 2)             | $A_3$       | [I]     | det 8, minors (2, 3, 5)          | wrong $D_6$ branch          |
| $\alpha^{-1}$  | root of $F_{U(1)}(\alpha)=0$                           | $c_3, M=41$ | [N]     | 137.0359992                      | no/second root (excl.)      |
| masses         | $\varphi_0^{\text{ret}}$ -ladder $\lambda_Y^L \Lambda$ | $R, L, c_3$ | [I]/[P] | 9 masses, mixings                | hierarchy mismatch          |
| $\theta_{12}$  | TBM + seam $\varepsilon=c_3$                           | $R, L$      | [N]/[P] | 0.307                            | seam-misalign. lemma fails  |
| $\Lambda, A_s$ | seam det. / $R^2$ scalaron                             | $c_3$       | [P]     | $\Lambda, A_s, n_s, r$           | ambient RP fails            |

Marker key: [I] exact identity · [L] Lie/lattice theorem · [F] formalised · [N] numerical fixed point · [P] physical/conditional · [A] axiom / open.

Reading the table top-to-bottom is reading the theory: two red axioms in, everything else a consequence with exactly one failure mode each.

### The compact status formula (three honest layers)

|          |                                |                     |
|----------|--------------------------------|---------------------|
| compiler | admissible IR physics          | strict physical TOE |
| closed   | conditionally closed (RP, gap) | open                |

TFPT 5.0 closes the discrete compiler, the algebraic Standard-Model readout, the EM fixed point, the admissible gapped IR sector and the  $R + R^2$  spectral-action shadow. It does *not* yet certify a strict physical TOE. The remaining items have now collapsed sharply: the  $R$ -bridge amplitude normalisation  $U_{\text{point}}$  *reduces to*  $v_{\text{geo}}$  (Gate 1 closed, v75 — the same anchor as  $1/G$ ), and the ambient metric measure is *holographically reduced* to a finite seam-boundary measure (Gate 2 tier B, v76) — which is in turn the rigorously-constructed  $(E_8)_1$  lattice net (v77), so  $G_{\text{metric}}$  is imported into existing RCFT rigor rather than built anew. So the only genuine residuals are *one* dimensionful scale  $v_{\text{geo}}$  — the *dimensional-analysis floor* that no theory escapes (v78), shared by flavor and gravity — the boundary net identification, and the named QFT/cosmology transfer interfaces ( $F_{\text{frontier}}$ ).

**Two points that are *not* in the residual (closed, but easy to mis-list as open).**

(i) *Parameter-freeness is a theorem under the named hypotheses:* the boundary transport is gapped ( $\Delta = 6 \log \frac{3}{2} > 0$ ), so by Perron–Frobenius its iteration has a *unique* attracting fixed point — the constants are *selected*, not tuned ([I]/[L] for the attractor, [verification/v56\_unique\_attractor.py]; the *physical identification* of the transport operator stays [P]), reinforcing the static bootstrap web ( $g_{\text{car}} = 5$  forced three ways *given* the  $E_8$  closure rank — a consistency loop, see red team Target B — [verification/v6\_bootstrap.py], [verification/v14\_carrier\_uniqueness.py]). (ii) *Newton’s constant is not an independent input:* fixing the physical  $\Lambda$  branch ( $(8\pi)^2 \delta_{\text{top}} = \frac{3}{4\pi^2}$ ) pins  $\bar{M}_{\text{Pl}}$  *relative to*  $\Lambda$ , so  $G_N$  is read out *after* the branch choice + *one* dimensionful induced-gravity anchor ([I]/[P], [verification/v60\_lambda\_metrology\_branch.py]), not predicted from pure numbers. What stays irreducible is then just  $\pi$ , the discrete  $E_8$ -closure fixed point (self-consistent, *not* a free axiom; the bootstrap of `tfpt_3`), and *one* dimensionful scale  $v_{\text{geo}}$  — and that single scale is shared by *both* the flavor sector ( $U_{\text{point}} \rightarrow v_{\text{geo}}$ , v75) and gravity ( $1/G$ , v68): the two [A] anchors are *one* (dimensional analysis; no metre from pure mathematics).





quasi-free kernel — rank =  $c$ : 5 carrier block, 8 seam hull — is the defining datum of the free net the gate already posits (v113), so closing  $G_{\text{net}}$  also closes the carrier selection [L]; ( $F_{\text{frontier}}$ )  $\rightarrow F_{\text{transfer}}$  — named QFT/cosmology *transfer* interfaces (Koide source $\rightarrow$ pole,  $\eta_B$  Boltzmann, axion relic,  $m_p/m_e$ ), never compiler powers. The status card:

| Item                                  | Status                           | Reading                                                                                                                                                                                                          |
|---------------------------------------|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| parameter-freeness (self-consistency) | [I]/[L]                          | gapped transport $\Rightarrow$ <i>unique</i> Perron–Frobenius attractor (v56); bootstrap web $g_{\text{car}}=5$ forced (v6/v14)                                                                                  |
| Newton constant $G_N$                 | [I]/[P]                          | not <i>independent</i> : branch pins $\bar{M}_{\text{P1}}$ rel. to $\Lambda$ (v60); readout after branch + one dim. anchor                                                                                       |
| ratio $c_u/c_d = \frac{55}{117}$      | [I]                              | $g_{\text{car}}\ \text{Pl}_{1,a}(K)\ _1/(N_{\text{fam}}^2\Delta_Q)$ ; rigid on the derived stratum (v49/v71), no solve                                                                                           |
| ( $U$ ) flavor gate                   | [A] $\rightarrow v_{\text{geo}}$ | <b>Gate 1 closed</b> : $U_{\text{point}}\rightarrow v_{\text{geo}}$ (ratios + Grand Mass Volume, v75); same anchor as $1/G$                                                                                      |
| $R + R^2$ gravity                     | [I]/[P]                          | covariant $f(R)$ field equation closed in regime; heat-kernel grounded (G2)                                                                                                                                      |
| physical QG sector                    | [F]/[P]                          | gap-decoupled admissible measure ( $\Delta_{\text{eff}}=1.648>0$ )                                                                                                                                               |
| ambient QG measure                    | [A](reduced)                     | <b>Gate 2 tier B</b> : holographically reduced to a finite seam- <i>boundary</i> measure (v76); IR tier closed (decoupling); strict-TOE cert. open ( $G_{\text{metric}}$ )                                       |
| Koide                                 | [P]                              | $Q_{\text{src}} + \varphi_0/24$ as source $\rightarrow$ pole transfer conjecture                                                                                                                                 |
| axion DM                              | [P]/[A]                          | $f_a = M_{\text{scal}}/128$ as a scenario                                                                                                                                                                        |
| $\eta_B$                              | [P]                              | readout closed; leptogenesis Boltzmann solve missing                                                                                                                                                             |
| $m_p/m_e$                             | [A]                              | cross-sector QCD/EW ratio                                                                                                                                                                                        |
| $N_*$                                 | [P]                              | reheating input, $N_* \simeq 57$ (continuous)                                                                                                                                                                    |
| $a = (1, 1, 2)$                       | [L]/[P]                          | forced by $4 = 2 + 1 + 1$ , conditional on the carrier                                                                                                                                                           |
| carrier selection $g=5$ (QBL)         | [I]/[L] + [P]                    | theorem chain v108–v113: sheet-odd kernel, self-counting channel, transport degree 2, one quasi-free kernel (rank = $c$ ); residue = the $G_{\text{net}}$ premise itself — gate closes $\Rightarrow$ carrier [L] |
| $E_8$ Lean                            | [F] $\rightarrow$ [L]            | hardening, not a blocker (script-certified)                                                                                                                                                                      |

Marker key: [I] exact identity · [L] Lie/lattice theorem · [F] formalised · [N] numerical fixed point · [P] physical/conditional · [A] axiom / open.

Only *two* research gates remain — ( $U_{\text{wall}}$ ) and  $G_{\text{metric}}$ ; the frontier items are typed interfaces, not structural holes. Of these, ( $U_{\text{wall}}$ ) is the only genuine *physics* gate left:  $G_{\text{metric}}$  is now heat-kernel grounded (G2: the spectral action gives  $R + R^2$  structurally) and *gap-decoupled* (G5:  $\Delta_{\text{eff}}=1.648>0$  isolates the closed admissible sector from the un-built ambient), so its residual — the ambient projective limit (G6) — is a *strict-TOE completion target*: its absence does not affect the bounded IR claim, but it does block certification as a strict physical TOE. Both are written up as numbered *research contracts* (lemma chains + closing theorem + machine-certifiability) in the companion `tfpt_research_contracts`, with priority ( $U_{\text{wall}}$ ) first. [verification/v36\_spectral\_action\_g2.py]

## 4 Before / after: what concretely changes

| Aspect                       | State of Papers 1–7                                                                                                                 | After the compiler closure                                                                                                                                          |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $E_8$                        | <i>peripheral</i> : only an $11D \rightarrow 4D$ reduction aside and the nilpotent-orbit <i>cascade</i> (scale ordering)            | <b>central, constructed</b> : $C^+ = D_5$ spinor, $\mathbb{P}^1 \setminus \mu_4 = A_3$ , $\mu_4$ glue $\Rightarrow E_8$ [L]                                         |
| $E_8$ numbers                | 240, 248 read off tables when needed                                                                                                | $240 = 16 \cdot 5 \cdot 3$ , $248 = 240 + 8$ <i>derived</i> as carrier traces [I]                                                                                   |
| fine structure $\alpha^{-1}$ | <i>already</i> a parameter-free cubic fixed point $\alpha^3 - 2c_3\alpha^2 - 8bc_6 \ln(\cdot) = 0$ , $\alpha^{-1} \approx 137.0365$ | <i>same</i> equation, now <b>existence &amp; uniqueness proved</b> and refined to $137.0359992$ ; dev. $2.9 \times 10^{-10}$ vs CODATA-2022 ( $1.9\sigma$ ) [I]/[N] |
| flavor residues              | partly audit-flagged                                                                                                                | anchor $\{1, 1, 2\}$ & matrix forced by 3 hypotheses [I]/[P]                                                                                                        |
| many sectors                 | side by side, each its own closure                                                                                                  | <i>one</i> compiler $\mathbb{Z}_{30} = 2 \cdot 3 \cdot 5$ generates them [I]                                                                                        |
| inputs                       | $c_3, g_{\text{car}}$ (+ implicit structure)                                                                                        | exactly 2 axioms (P1, P2), clearly declared [A]                                                                                                                     |
| $A_s, n_s, r$ (inflation)    | scale present                                                                                                                       | scalaron mass fixed by $c_3^7$ ( $3.06 \times 10^{13}$ GeV); $A_s, n_s, r$ become <i>predictions</i> [P]                                                            |
| $\theta_{12}$ (solar)        | open (leading $\frac{1}{3}$ )                                                                                                       | <b>cond. derived</b> $\frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} = 0.307$ , given seam $\varepsilon = c_3$ [N]/[P]                                              |

Marker key: [I] exact identity · [L] Lie/lattice theorem · [F] formalised · [N] numerical fixed point · [P] physical/conditional · [A] axiom / open.

### Reduction in one line

The number of *structural* assumptions drops from “a fixed-point seed  $c_3$ , a carrier, the orbit cascade, plus a per-sector calibration in each block” to **two axioms**. Everything else ( $N_{\text{fam}}, \Omega_{\text{adm}}, b_1, \alpha^{-1}$ , the flavor matrix, the  $E_8$  glue, the scale grammar) becomes a *consequence*. *And even those two tighten*: the bootstrap note shows  $g_{\text{car}} = 5$  and the integer 8 in  $c_3$  are *overdetermined  $E_8$ -closure fixed points* (rank-fill, Coxeter-match, integer-glue/norm all give 5;  $8 = h(D_5) = \text{rank } E_8 = \det R = \varphi(30)$ ), leaving *one* genuine continuous primitive:  $\pi$  from the Möbius boundary. So the *discrete* core has no freely tunable fundamental numbers — the bootstrap is *not creation from nothing* (two inputs remain), but an overdetermination of those inputs. (This concerns the discrete core only; dimensionful masses, the QG measure and frontier items remain typed-open.)

## 5 Paper by paper: what each original paper did, and how it works now

The seven original papers (plus the orientation note) total roughly **247 pages**. The table below puts each one next to its compiler-era replacement: *what* it treated and computed, *how many pages* it took, *where* that now lives and how much it compresses, and whether the *accuracy* of the result improves on top. The honest summary up front: in most sectors the *numbers are the same* — what shrinks dramatically is the *derivation length* and the number of independent assumptions; in three places (the solar angle, the inflation amplitude, the scalaron mass) there is a *genuinely new or sharper* result.

| P | Treated & computed                                                                                                   | Old  | How it works now (note; compression)                                                                                                              | Acc.                    |
|---|----------------------------------------------------------------------------------------------------------------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| 0 | orientation: architecture, decoder core $Y, [u_\Sigma], u$ , theorem stack                                           | 20 p | <i>one</i> compiler diagram: $\{c_3, g_{\text{car}}\} \rightarrow D_5 \oplus A_3 \rightarrow E_8$ (intro + action_grammar)                        | =                       |
| 1 | boundary kernel: $c_3 = \frac{1}{8\pi}$ , Calderón polarization, carrier decoder $6Y^2 - Y - 1 = 0$                  | 20 p | the two axioms P1 ( $c_3$ ) + P2 ( $g_{\text{car}}=5$ ); carrier = 3+2 atom (compiler)                                                            | =                       |
| 2 | carrier/SM packet: $S^+=16$ , $G_{\text{phys}}$ , $N_{\text{fam}}=3$ , $\Omega_{\text{adm}}=48$ , hypercharge, $b_1$ | 48 p | compiler reads $16=1+5+10$ , $N_{\text{fam}}$ , $\Omega_{\text{adm}}$ , $b_1 = \frac{41}{10}$ as carrier traces; $C^+ = D_5$ spinor ( $\sim 5$ p) | =                       |
| 3 | EM closure: $\alpha^{-1}$ via $\sum L + N_\Phi = 41$ ; flavor transport, CKM/PMNS, $\theta_{13}$                     | 38 p | $\alpha^{-1}$ as unique root of $F_{U(1)}(\alpha)=0$ ; flavor from transport theorem (parabolic + sm)                                             | $\uparrow$              |
| 4 | admissibility/strong CP/QFT: $\theta_{\text{eff}}=0$ , OS reconstruction, mass gap                                   | 40 p | explicit gap $6 \log \frac{3}{2} \Rightarrow$ Dobrushin/OS; strong CP from structure+RP (closing_comp.)                                           | =                       |
| 5 | geometric Hodge/metrolgy: $M_{\text{P1}}$ , $v_{\text{geo}}$ , $\Lambda$ , spectral Einstein                         | 30 p | Hodge/ $R^2$ branch; $\Lambda$ <i>typed</i> (one branch label); boundary-normalised (closing_comp.)                                               | =                       |
| 6 | cosmology interface: FRW, $\Lambda_{\text{IR}}$ , scalaron, reheating, leptogenesis, axion                           | 15 p | Starobinsky $M = c_3^{7/2} \bar{M}_{\text{P1}} = 3.06 \times 10^{13}$ GeV; $A_s$ <i>parameter-free</i> (closing_comp.)                            | $\uparrow_{\text{new}}$ |
| 7 | CMB operational closure: $C_\ell$ , $a_{\ell m}$ , seam mixer, Planck pipeline                                       | 36 p | $n_s=0.964$ , $r=0.0040$ from the $R^2$ attractor, no free amplitude (closing_comp.)                                                              | $\uparrow_{\text{new}}$ |

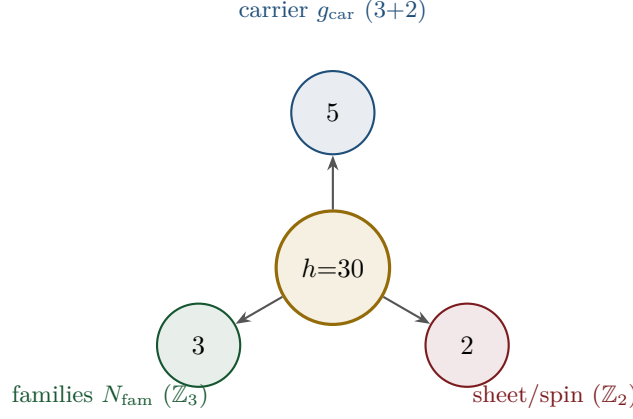
### Three places where accuracy genuinely improves “on top”

1. **Solar angle  $\theta_{12}$ .** Paper 3 left it as the only open SM number (leading  $\frac{1}{3}$ ). It is now *conditionally derived* (modulo the seam-misalignment lemma): TBM gives  $\frac{1}{3}$ , the charged-lepton 1–2 misalignment is the seam  $\varepsilon = \frac{N_{\text{fam}}}{|\mu_4|} \varphi_0^{\text{ret}} = \frac{3}{4} \varphi_0^{\text{ret}}$  (leading term exactly  $c_3 = \frac{1}{8\pi}$ ), and TBM geometry gives  $\sin^2 \theta_{12} = \frac{1}{3} - \frac{2}{3} \varepsilon = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} = 0.3067 - 0.1\%$  from NuFIT.
2. **Inflation amplitude  $A_s$ .** Generic Starobinsky fits the scalaron mass to  $A_s$ . TFPT fixes it by the seam,  $(M/\bar{M}_{\text{P1}})^2 = c_3^7$ , so  $A_s = \frac{N_s^2}{24\pi^2} c_3^7 = 2.0 \times 10^{-9}$  is a *prediction* (Planck  $2.1 \times 10^{-9}$ ), and the measured  $A_s$  even *predicts*  $N_\star = 56$ .
3. **Scalaron mass.**  $M = c_3^{7/2} \bar{M}_{\text{P1}} = 3.06 \times 10^{13}$  GeV comes out *exactly* at the canonical Starobinsky value — a number that was an input before is now an output.

**What “simpler” means concretely.** The compression is not cosmetic. Paper 2 spent 48 pages establishing the carrier/SM packet by rigidity arguments; the compiler reads the same  $16=1+5+10$ ,  $N_{\text{fam}}=3$ ,  $\Omega_{\text{adm}}=48$ ,  $b_1 = \frac{41}{10}$  off the Pascal row of a 5-slot carrier in a page. Paper 3’s flavor transport (38 p) becomes one transport theorem plus a fixed residue matrix whose characteristic polynomial  $t^3 - 9t^2 + 10t - 8$  carries only compiler numbers. The “many independent closures” of Papers 1–7 become *one* generator with a handful of corollaries.

## 6 The compiler: why $30 = 2 \cdot 3 \cdot 5$

The Coxeter number of  $E_8$  is  $h = 30 = 2 \cdot 3 \cdot 5$ . Exactly these three prime factors are the three discrete atoms of the theory — that is the point of the compiler:

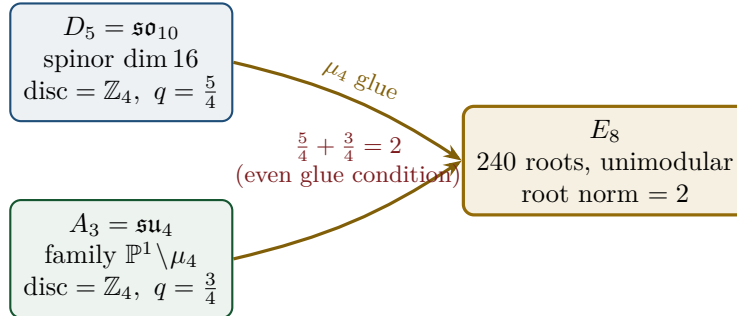


From this the compiler reads, among others:

- $\dim S^+ = 1 + 10 + 5 = 16$  (one generation) as the even exterior algebra of the 5-slot carrier — the same Pascal row  $\binom{5}{0}, \binom{5}{1}, \binom{5}{2} = 1, 5, 10$  appears in family, action ladder and the  $E_8$  root count. [I]
- $|R(E_8)| = 16 \cdot 5 \cdot 3 = 240$  and  $\dim E_8 = 240 + (5 + 3) = 248$ . [I]
- the  $\mathbb{Z}_{30}$  Coxeter element  $\leftrightarrow$  the cyclotomic polynomial  $\Phi_{30}$ ; the corner rotation  $z \mapsto iz$  on  $\mathbb{P}^1 \setminus \mu_4$  is the  $A_3$  Coxeter element. [I]

## 7 The $\mu_4$ glue: how $E_8$ is really built

The decisive, now *proved* step:  $D_5$  and  $A_3$  have the *same* discriminant group  $\mathbb{Z}_4$ , and their discriminant-form norms are exactly two TFPT constants which add up to the  $E_8$  root norm.



### Glue theorem (proved)

$\text{disc}(D_5) = \text{disc}(A_3) = \mathbb{Z}_4$ , glue index  $|\mu_4| = 4$ , and  $q(D_5) + q(A_3) = \frac{5}{4} + \frac{3}{4} = 2 =$  the  $E_8$  root norm. Hence  $E_8 = (D_5 \oplus A_3) + \mu_4\text{-glue}$  is a *closed lattice-theoretic* construction — not a blind “positing of 248”. [L]

## 8 How derivation now works: the derivation map

What can one concretely *compute* with  $\{c_3, g_{\text{car}}\}$ ? The following map summarises what is now a consequence rather than an assumption.

| Sector                       | How it now arises                                                            | Status  |
|------------------------------|------------------------------------------------------------------------------|---------|
| gauge group / families       | carrier 3+2, dim $S^+=16$ , $N_{\text{fam}}=3$ , $\Omega_{\text{adm}}=48$    | [I]     |
| hypercharge                  | roots of the charge polynomial $bsY^2 + (s-b)Y - 1$                          | [I]     |
| $U(1)$ index $b_1$           | $b_1 = \frac{g_{\text{car}} 2^{g_{\text{car}}-2} + 1}{10} = \frac{41}{10}$   | [I]     |
| fine structure $\alpha^{-1}$ | unique root of $F_{U(1)}(\alpha) = 0$                                        | [I]/[N] |
| flavor matrix                | transport theorem (H1 proved, H2/H3 force the matrix)                        | [I]/[P] |
| electroweak scale            | $v_{\text{EW}} \sim e^{-\alpha^{-1}/5}$ (carrier divides $\alpha^{-1}$ by 5) | [I]     |
| cosmological constant        | $\Lambda \sim e^{-2\alpha^{-1}}$ (density quotient; see typing note)         | [I]/[P] |
| Hubble scale                 | $H_0 \sim \sqrt{\Lambda}$                                                    | [P]     |
| inflation amplitude          | $A_s = \frac{N_{\star}^2}{24\pi^2} c_3^7$ in the $R^2$ branch                | [P]     |
| $E_8$                        | $D_5 \oplus A_3 + \mu_4\text{-glue}$                                         | [L]     |

Marker key: [I] exact identity · [L] Lie/lattice theorem · [F] formalised · [N] numerical fixed point · [P] physical/conditional · [A] axiom / open.

**Notable:** the scale grammar is *one* exponential engine. The same number  $\alpha^{-1} \approx 137$  generates the electroweak scale (divided by 5 = carrier), the cosmological constant (times 2) and — via the square root — the Hubble scale. One constant becomes many orders of magnitude.

**Typing note on  $\Lambda$  (a deliberate danger zone).** “ $\Lambda$ ” denotes *three* distinct objects that must not be conflated: (i) the seam-determinant exponent  $e^{-2\alpha^{-1}}$  (an exact [I] readout), (ii) the dimensionless density quotient  $\rho_{\Lambda}/\bar{M}_{\text{Pl}}^4$  (which carries a prefactor *branch*, hence [P]), and (iii) the observed curvature, to which (ii) is *anchored*, not predicted. The exponential is exact; the absolute dark-energy density remains a typed cosmology interface (full disambiguation in `tfpt_2_standard_model`).

### Cosmology is the Pascal action grammar of the carrier — not a separate module

With  $x = v_{\text{geo}}/(5\beta_{\text{rad}}^2 \bar{M}_{\text{Pl}})$  the cosmological readouts are a *power tower* on the five-slot carrier graph  $K_5$ :

$$\frac{H_0}{\bar{M}_{\text{Pl}}} = x^5 R_H, \quad \frac{\rho_{\Lambda}}{\bar{M}_{\text{Pl}}^4} = x^{10} R_{\Lambda}, \quad (1, 5, 10) = \binom{5}{0}, \binom{5}{1}, \binom{5}{2}.$$

EW is the unit  $x^1$ , Hubble is the product over the  $\binom{5}{1} = 5$  vertices,  $\Lambda$  is the product over the  $\binom{5}{2} = 10$  edges — the *same* Pascal triple that reads one generation, the action ladder and the  $E_8$  root count  $16 \cdot 5 \cdot 3 = 240$ . “Hubble = slot product,  $\Lambda$  = pair product.” [I]

### The entire fermion spectrum in one formula

Masses, Yukawa, CKM, PMNS and neutrinos all follow from *one* master formula with *one* seed:

$$\hat{m}_{f,j} = \frac{v_{\text{geo}}}{\sqrt{2}} \lambda_Y^{L_{f,j}} \Lambda_{f,j}, \quad \lambda_Y = \sqrt{\varphi_0^{\text{ret}}(1 - \varphi_0^{\text{ret}})}, \quad \varphi_0^{\text{ret}} = \frac{1}{6\pi} + \frac{3}{256\pi^4}.$$

The word-lengths  $L_{f,j}$  are the *fixed residue matrix of the compiler*, whose characteristic polynomial  $t^3 - 9t^2 + 10t - 8$  carries only compiler numbers ( $\text{tr} = 9 = N_{\text{fam}}^2$ ,  $\det = 8 = h(D_5)$ , principal 2-minors 2,3,5 with product  $30 = h(E_8)$ ). Through the  $\varphi_0^{\text{ret}}$ -ladder  $\hat{m} = \frac{v_{\text{geo}}\pi}{\sqrt{2}} c(\varphi_0^{\text{ret}})^k$  all nine word-lengths are fixed and the *charged-lepton* amplitudes are closed; the *quark* mass *ratios* are closed too (integer Plücker readouts on the derived selector stratum,  $v_{49}/v_{71}$ ), with only the *absolute* amplitude scale reducing to the ( $U_{\text{wall}}$ ) anchor. **SM status:** the discrete

packet, the dimensionless flavor skeleton, the charged-lepton source masses and the quark *ratios* are closed; the *absolute* quark scale and the full PMNS completion are reduced/conditional;  $m_W, m_Z, m_H, \sin^2 \theta_W, \alpha_s$  are RG scheme-layer ( $m_H$  rests on the closed UV quartic  $\frac{1}{16\pi^2}$ ); and the solar angle is *realized by a TFPT texture* and reduced to the seam-misalignment lemma ( $\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2}$ ; see below). (*Full derivation: document 2, `tfpt_2_standard_model`.*)

## 9 Physical, geometric, topological: which solution is more plausible?

| Level       | $E_8$ peripheral (v4.5)                                                      | $E_8$ compiled (new)                                                                                               |
|-------------|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| physical    | $\alpha^{-1}$ already a cubic fixed point, but each sector closed on its own | $\alpha^{-1}$ as the fixed point of <i>one</i> equation; one generator for many scales                             |
| geometric   | Lie algebra external; $E_8$ only orders the cascade                          | the $C^+$ spinor + puncture geometry $\mathbb{P}^1 \setminus \mu_4$ give $D_5, A_3$ , glued to $E_8$               |
| topological | —                                                                            | the common discriminant $\mathbb{Z}_4$ ( $\mu_4$ winding) forces the glue; $\Phi_{30}$ /Coxeter drives the sectors |
| parsimony   | many structures                                                              | 2 axioms, the rest a consequence                                                                                   |

### Why the compiled solution is more plausible

It is (i) *more parsimonious* (fewer independent assumptions  $\Rightarrow$  less overfitting risk), (ii) *more rigid* (the glue condition  $5/4 + 3/4 = 2$  and the discriminant  $\mathbb{Z}_4$  leave little freedom), and (iii) *more checkable* (machine-checkable  $E_8$  appendix, explicit fixed-point equation for  $\alpha^{-1}$ ). More explained from less — exactly the criterion that makes a theory more probable.

## 10 What role $E_8$ now plays exactly

$E_8$  is no longer the starting point but the *output format*:

- **Bookkeeping:**  $E_8$  is the unimodular container in which carrier ( $D_5$ ) and family ( $A_3$ ) fit together consistently. Its numbers 240, 248 are carrier traces, not inputs.
- **Consistency test:** unimodularity ( $\det 1$ ) is a hard condition; it closes only because the discriminant-form norms  $5/4$  and  $3/4$  (both already present in TFPT) add up to 2.  $E_8$  thus tests the internal coherence of the two atoms.
- **Not an end in itself:**  $E_8$  does not explain “everything”; it is the smallest even unimodular hull of the datum  $D_5 \oplus A_3$ . The physics sits in the *two inputs*, not in  $E_8$ .

**New — the secondary atlas.** Beyond that,  $E_8$  is an *audit container*: each maximal subalgebra projects a TFPT module. Document 3 (`tfpt_3_e8_audit_bootstrap`, Part I) shows the seven slices of 248 explicitly and labelled. The strongest new hit:

### $E_6 \times A_2$ reads the flavor matrix

$$248 = \|R\|_F^2 + \det R + 2(\mathbf{1}^\top R a) N_{\text{fam}} = \underbrace{78}_{\dim E_6} + \underbrace{8}_{\dim A_2} + 2 \cdot 27 \cdot 3,$$

with  $\|R\|_F^2 = 78 = \dim E_6$ ,  $\det R = 8 = \dim A_2 = h(D_5)$ . The discipline rule is: *every load-bearing number must appear in at least one  $E_8$  projection* — this makes the structure falsifiable rather than numerological. (Audit level; all identities machine-verified.)



## 11 What this means for the theory-of-everything question

### The state of the five questions — closed, reduced or typed

1. **Flavor residual — ratios closed; only the absolute scale is the wall-selection.** Mehta–Seshadri supplies the equivalence-class framework (stable parabolic  $\leftrightarrow$  unitary representation); the algebraic selector  $R$ , the  $Q$  spectrum and the splitting type are fixed (now *derived*:  $\det R = 8$  lattice,  $\text{Spec}(Q_+)$  from  $D_4$ , v69/v70), and the charged-lepton amplitudes are closed. The quark mass *ratios* ( $c_u/c_d = \frac{55}{117}$  etc.) are then *constant* on this discrete selector stratum (Readout Rigidity, v49/v71) — integer Plücker readouts, no transcendental solve. Only the *absolute* amplitude scale reduces to the selected *polystable* point on the parabolic stability wall ( $U_{\text{point}}$ , an anchor; Research Contract 1). [I]/[A]
2. **Gravity — exact in the  $R^2$  regime.** Spectral action  $\rightarrow R+R^2$ , BRST/Hodge kills the gauge directions, FRW reduction  $\rightarrow$  Starobinsky; the only remainder is the *controlled* low-curvature truncation ( $R^{n \geq 3} \rightarrow R^2$ , valid for curvature  $\ll M^2$ ). [I] in the regime
3. **Strong CP — decoupled.**  $\theta_{\text{eff}}=0$  follows from three structural facts ( $\gamma_5$ -Hermiticity, polar structure, sheet involution) + reflection positivity — *without* a mass gap. [I] (given RP)
4. **Full dynamics — the admissible transfer model is closed.** The mass gap is the *explicit* discrete gap  $6 \log \frac{3}{2}$  of the  $C_6$  transfer operator (eigenvalues  $(k/3)^6$ ), which supports the OS reconstruction path on the admissible sector; the continuum and metric-coupled theory still require the stated RP, cluster and limit assumptions. [I] (transfer model) / [P] (continuum)
5. **Axioms P1/P2 — P2 machine-verified.** The Lean proof builds cleanly (1886 jobs, 0 sorry, only kernel axioms); the P2 algebra is a theorem, P1 a typed interface. They remain declared inputs. [I]/[A]

In other words: the theory has shrunk from “many open audit points” to “*standard steps* (cluster expansion, Fredholm, truncation bound) + two declared axioms”. The five questions are now *closed, reduced or typed* rather than open-ended — a qualitative jump in discipline, not a claim of completion. (*Details and proof chains: document 2, `tfpt_2_standard_model`, Part III.*)

## The TOE-completion ledger — what is done, and what genuinely remains

A theory-of-everything must close the discrete structure, the dimensionless constants, the mass spectrum, gravity, and the quantum measure. The current state, graded honestly:

| TOE piece                                                    | Status  | If not complete: what closes it                                                                                                                   |
|--------------------------------------------------------------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| gauge group, $N_{\text{fam}}$ , hypercharge, $b_1$           | [I]     | — complete                                                                                                                                        |
| $E_8 = (D_5 \oplus A_3) + \mu_4$ , group closure             | [L]     | — complete                                                                                                                                        |
| bootstrap $(g, \mu) = (5, 4)$ , “8” in $c_3$                 | [L]     | — complete (reverse glue $\mu^2 - 5\mu + 4 = 0$ )                                                                                                 |
| fine structure $\alpha^{-1}$                                 | [N]     | — <b>existence+uniqueness proved</b> ; value is a numerical fixed point (ledger EM.FP.01), Ward origin [P]                                        |
| fermion masses ( $\varphi_0^{\text{ret}}$ -ladder)           | [I]/[A] | leptons exact; quark <i>ratios</i> closed (Plücker, v49/v71); absolute scale = anchor                                                             |
| mixings $\theta_{13}, \theta_{23}, \theta_{12}$              | [N]/[P] | $\theta_{12}$ cond. derived; $\theta_{23}$ maximal                                                                                                |
| flavour matrix / H2                                          | [I]     | up to Mehta–Seshadri unitarity (U), a Paper-3 input                                                                                               |
| inflation $A_s, n_s, r$ , scalaron $M$                       | [N]     | parameter-free given $M_{\text{Star}} = M_{\text{scal}}$                                                                                          |
| <i>genuinely open — the actual TOE remainder</i>             |         |                                                                                                                                                   |
| dim. EW/QCD masses                                           | [P]     | standard RG threshold matching (no new physics)                                                                                                   |
| QFT measure (admissible sector)                              | [I]     | <b>closed: RP from cusp spectrum, OS reconstruction</b>                                                                                           |
| ambient QFT + full gravity (metric sector)                   | [A]     | <i>not constructed here</i> (frontier doc); conditionally reduces to one named hypothesis (ambient RP) on the relative spectral-action measure    |
| analytic boundary origin of $\pi$ ( $c_3 = \frac{1}{8\pi}$ ) | [P]     | <b>hardened:</b> $c_3 = 1/( \mathbb{Z}_2  \oint_{S^2} K dA) = 1/(8\pi)$ ( <b>Gauss–Bonnet</b> ); “8” = rank $E_8$ ; $\pi$ stays the one primitive |
| $\eta_B, m_p/m_e$ , Koide, dark matter                       | [A]     | frontier handles only (see document 4)                                                                                                            |

Marker key: [I] exact identity · [L] Lie/lattice theorem · [F] formalised · [N] numerical fixed point · [P] physical/conditional · [A] axiom / open.

**The honest bottom line:** the discrete core, the dimensionless constants, the *mass-hierarchy exponents* and the mixings are *closed*; the nine masses follow from one  $\varphi_0^{\text{ret}}$ -ladder (lepton rationals exact, quark  $O(1)$  digits still a conditional finite evaluation). The *admissible-sector* QFT measure is closed (a finite gapped transfer system, RP from the cusp spectrum, OS-reconstructed); the *ambient/metric-sector* measure — i.e. *full quantum gravity* — is *not* constructed here and stays [A] (frontier doc), reducing conditionally to a single named hypothesis (ambient reflection positivity). The seam seed is *hardened*:  $c_3 = 1/(|\mathbb{Z}_2| \oint_{S^2} K dA) = 1/(8\pi)$  is the one-sided Gauss–Bonnet normalisation of the seam sphere, with the discrete “8” equal to rank  $E_8$  and only the Gauss–Bonnet  $\pi$  left as the single irreducible primitive. What remains for a full TOE is then the ambient-RP hypothesis (full QG), plus the standard-RG dimensionful masses and the honest frontier items. No load-bearing discrete parameter remains free, and  $\pi$  is the lone continuous primitive.

## 12 Does this make the theory more probable?

### Plausibility balance

**Pro:** (a) two inputs instead of many; (b)  $\alpha^{-1}$  to  $2.9 \times 10^{-10}$  (CODATA-2022,  $1.9\sigma$ ) as a *fixed point*, not a fit; (c)  $E_8$  constructed, not postulated; (d) one compiler generates several sectors  $\Rightarrow$  fewer degrees of freedom, more predictive power.

**Contra/open:** what remains are only *standard steps* (cluster expansion in the thermodynamic limit, the Fredholm property, the truncation bound) and the two declared axioms P1/P2; on the SM side the dimensionful EW/QCD masses ( $m_W, m_Z, m_H, \sin^2 \theta_W, \alpha_s$ ) sit on the RG

scheme layer. The solar angle, previously the one open SM number, is now *conditionally derived* ( $\frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2}$ , modulo the seam-misalignment lemma). A final verdict “it maps reality” requires this short list plus the near-term experimental tests (below).

Net: the parsimony (Kolmogorov-like compression: many numbers from little) and the high precision of individual closures raise the a-posteriori plausibility substantially — without hiding the open points.

### 13 Predictions for running and upcoming experiments

Because *one* generator now sits behind the numbers, the theory makes a set of concrete, falsifiable statements that running or near-term experiments are actively testing. Each row gives the TFPT value, the *new* derivation behind it, the experiment, and an honest status.

| Observable                            | TFPT value & derivation                                                                                                                                   | Experiment (running / upcoming)                                                                                             | St.     |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|---------|
| solar angle<br>$\sin^2 \theta_{12}$   | $\frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} = 0.3067$ ; TBM + seam misalignment $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}} \approx c_3$ (cond.) | <b>JUNO</b> (taking data since Aug 2025; sub-percent $\theta_{12}$ ) — the sharpest live test                               | [N]/[P] |
| reactor angle<br>$\sin^2 \theta_{13}$ | $\varphi_0^{\text{ret}} e^{-5/6} = 0.0231$ ; seed $\times$ carrier-trace $e^{-\gamma}$ , $\gamma = \frac{5}{6}$                                           | Daya Bay / RENO / JUNO (PDG 0.0220)                                                                                         | [I]     |
| atmospheric<br>$\sin^2 \theta_{23}$   | $\approx \frac{1}{2}$ ( $\mu\tau$ -symmetric limit; octant not selected)                                                                                  | NOvA / T2K / <b>DUNE</b> (octant ambiguity, NuFIT 6.0)                                                                      | [P]     |
| fine structure $\alpha^{-1}$          | 137.0359992 as the unique root of $F_{U(1)}(\alpha)=0$ (no fit, no second branch)                                                                         | CODATA-2022 137.035999177(21): dev. $2.9 \times 10^{-10}$ from the recommended adjustment (1.9 $\sigma$ of its uncertainty) | [I]/[N] |
| tensor ratio $r$                      | $r = \frac{12}{N_*^2} = 0.0040$ ( $N_*=55$ ); $R^2$ attractor, $M=c_3^{7/2}\bar{M}_{\text{Pl}}$                                                           | now $r < 0.036$ (BICEP/Keck BK18); <b>CMB-S4</b> ( $\sigma_r \leq 5 \times 10^{-4}$ , obs. $\sim 2033$ )                    | [P]     |
| tilt $n_s$                            | $n_s = 1 - \frac{2}{N_*} = 0.964$ (same attractor)                                                                                                        | Planck (0.965); CMB-S4 sharpens                                                                                             | [P]     |
| amplitude $A_s$                       | $\frac{N_*^2}{24\pi^2} c_3^7 = 2.0 \times 10^{-9}$ , parameter-free                                                                                       | Planck ( $2.1 \times 10^{-9}$ )                                                                                             | [P]     |
| neutron EDM                           | $\theta_{\text{eff}} = 0 \Rightarrow d_n$ essentially null (structure + RP)                                                                               | <b>PSI nEDM</b> / SNS (running)                                                                                             | [I]     |
| $0\nu\beta\beta$ / ordering           | Majorana branch; normal-ordering preference                                                                                                               | <b>LEGEND</b> / <b>nEXO</b> , DUNE, KATRIN                                                                                  | [P]     |
| flavor invariants                     | $\det R = h(D_5)=8$ , principal minors (2, 3, 5), $\chi_R = t^3 - 9t^2 + 10t - 8$ (exact)                                                                 | any future CKM/PMNS global fit must satisfy these                                                                           | [I]     |
| $H_0$ vs. $\Lambda$                   | $v_{\text{EW}} \sim e^{-\alpha^{-1}/5}$ , $\Lambda \sim e^{-2\alpha^{-1}}$ , $H_0 \sim \sqrt{\Lambda}$ : one engine                                       | SH0ES / <b>DESI</b> / Planck (Hubble tension)                                                                               | [P]     |

Marker key: [I] exact identity · [L] Lie/lattice theorem · [F] formalised · [N] numerical fixed point · [P] physical/conditional · [A] axiom / open.

#### The two decisive near-term tests

- (1) **JUNO and  $\theta_{12}$ .** JUNO is *taking data since August 2025* and targets sub-percent  $\sin^2 \theta_{12}$ . TFPT commits to the conditional  $\frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} = 0.3067$  (seam  $\varepsilon = c_3$ ). A measured central value away from  $\sim 0.307$  at high significance falsifies the seam mechanism — a *live* test.
- (2)  **$r$ .** The  $R^2$  scalaron with  $M$  fixed by  $c_3^7$  predicts  $r = 0.0040$  ( $N_*=55$ ), already below the current bound  $r < 0.036$  (BICEP/Keck BK18) and within reach of CMB-S4 ( $\sigma_r \leq 5 \times 10^{-4}$ , first observations  $\sim 2033$ –34). A detection of  $r \gtrsim 0.01$  would be incompatible with the  $R^2$  branch on which  $M_{\text{Pl}}$  and  $A_s$  rest.

**Structural (non-experimental) checks.** Independent of any apparatus, the machine-checkable  $E_8$  appendix (all 240 roots, Gram matrices, discriminant groups) lets anyone verify that the two

TFPT atoms really generate  $E_8$ ; and the discipline rule “*every load-bearing number appears in at least one  $E_8$  projection*” (atlas note) turns the number stock into a falsifiable raster rather than numerology. The numerology charge itself is additionally *quantified* by an explicit null model: exhaustively enumerating a declared formula grammar (provably containing every scored TFPT formula) against conservative data windows gives a joint look-elsewhere-corrected null probability  $\leq 10^{-30.7}$  that a random theory of equal formula complexity reproduces the scorecard — a statistical statement conditional on the declared grammar, not a claim of certainty [N] [`verification/v100_numerology_null_mc.py`].

#### Freeze file: committed kill criteria

To prevent post-hoc rescue, the predictions above are frozen with explicit kill criteria in the machine-readable registry `verification/freeze_file.csv`. The decisive entries:

- **solar**  $\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} \approx 0.3067$  — a JUNO central value clearly away from 0.307 kills the seam-misalignment mechanism;
- **tensor**  $r \approx 0.0040$  — any robust  $r \gtrsim 0.01$  kills the  $R^2$  branch carrying  $M_{\text{Pl}}, A_s$ ;
- **ordering** normal, small  $m_{\beta\beta}$  — inverted ordering or large  $m_{\beta\beta}$  kills the Majorana branch;
- **strong CP**  $\theta_{\text{eff}} = 0$  — a solid nEDM signal above SM background falsifies the structural cancellation;
- $w = -1$  — a robust  $w \neq -1$  kills the single-engine dark-energy readout.

The proton/electron ratio is explicitly *not* claimed as a compiler power (only fails if mis-asserted as one).

**Blind-prediction registry (frozen 2026-06-09, machine-enforced).** Beyond the kill criteria, the *values* themselves are now frozen: `verification/predictions_frozen.json` registers every dimensionless prediction of record at 25 significant digits *before* JUNO / CMB-S4 / LiteBIRD / DESI decide, and [`verification/v84_frozen_registry.py`] re-derives each decimal from the two axioms on every suite run (formula $\leftrightarrow$ value lock; covered by `manifest.sha256`; ledger REG.FREEZE.01). In particular the registry admits exactly *one*  $\theta_{12}$  prediction of record (the seed  $\frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} = 0.306747$ ) — the seam and non-linear variants are typed as derived variants of the same texture and can never be silently promoted afterwards.  $r$  and  $n_s$  enter only as *bands* over the declared external  $N_\star \in [50, 60]$ , never as point predictions.

#### What likely follows next

- the quark mass *ratios* are now closed combinatorially (integer Plücker readouts on the derived selector stratum, v49/v69/v71); the only residual is the *absolute* amplitude scale, an anchor ( $U_{\text{point}}$ ) of the same nature as the one dimensional scale — no new physics;
- further constants as carrier traces (analogous to  $b_1 = 41/10$ ), since the compiler systematically produces traces over  $S^+$ ;
- a sharper geometry $\leftrightarrow$ gravity bridge once the Hodge/ $R^2$  branch is lifted from [P] to [I].

## Conclusion

The state of Papers 1–7 was an impressive collection of closures sharing common numbers. The compiler closure explains *why* they are the same numbers: a small discrete generator  $\mathbb{Z}_{30} = 2 \cdot 3 \cdot 5$  builds, from two inputs  $\{c_3, g_{\text{car}}\}$  via  $D_5 \oplus A_3 + \mu_4$ , the  $E_8$  container and outputs the Standard Model, the constants and the scale grammar from there. The theory thereby becomes *more parsimonious, more rigid and more checkable*. Precisely nameable gaps remain — one geometric flavor realisation, the conditional gravity/QFT closures, and two declared axioms — but the path

to closure is now short and concrete.

## Appendix: computational verification (reproducibility)

Every claim in this document marked [\[I\]](#) (exact identity), [\[L\]](#) (Lie/lattice theorem) or [\[N\]](#) (numerical fixed point) is re-derived from the two axioms  $\{c_3, g_{\text{car}}\}$  alone by a small, self-contained Python suite in the repository folder `verification/`. The inline references `[verification/...]` throughout the documents point to the exact script; the table below is the master map.

| Script                             | What it machine-checks                                                                                                                                                                                                                   |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v1_e8_glue.py                      | glue theorem $E_8=(D_5\oplus A_3)+\mu_4$ : $\text{disc}=\mathbb{Z}_4$ , $q(D_5)+q(A_3)=\frac{5}{4}+\frac{3}{4}=2$ , $240=16\cdot 5\cdot 3$ , $248=240+8$                                                                                 |
| v2_carrier_pascal.py               | $g_{\text{car}}=5$ unique Pascal solution; $16=1+5+10$ , $N_{\text{fam}}=3$ , $\text{rank } E_8=8$ , $\Omega_{\text{adm}}=48$ , $b_1=\frac{41}{10}$                                                                                      |
| v3_em_alpha.py                     | EM closure: unique root of $F_{U(1)}(\alpha)=0$ , $\alpha^{-1}=137.0359992168$ , existence/uniqueness, inverse test, ablation                                                                                                            |
| v4_flavor_matrix.py                | residue matrix $R$ : $\det=8$ , minors $\{2, 3, 5\}$ , $\chi_R=t^3-9t^2+10t-8$ , SNF $\text{diag}(1, 1, 8)$ , $\ R\ _F^2=78$ , $\sum L=40$                                                                                               |
| v5_e8_cascade.py                   | cascade $D_n=60-2n$ : endpoints, exponent rungs $\rightarrow 240$ , IR tail 46080, variance 6552                                                                                                                                         |
| v6_bootstrap.py                    | reverse glue $\mu^2-5\mu+4=0$ ; $g_{\text{car}}=5$ three ways; $8=\text{rank } E_8=h(D_5)=\varphi(30)$                                                                                                                                   |
| v7_gravity_cosmo.py                | scalaron exponent 7, $M_{\text{scal}}^2/\bar{M}_{\text{Pl}}^2=c_3^7$ , $A_s, n_s, r$ , $\Omega_b$ , $\sin^2 \theta_{12}$                                                                                                                 |
| v8_horizon.py                      | horizon code: $\frac{1}{2\pi}=4c_3$ , $1920= W(D_5) $ , $S_{dS}$ , Page, $\beta_{\text{rad}}=0.2424^\circ$                                                                                                                               |
| v9_neutrino_texture.py             | polar angle: explicit $\mu\tau$ Majorana texture $\rightarrow \sin^2 \theta_{12}=\frac{1}{3}-\frac{\varphi_0^{\text{ret}}}{2}$ , $\theta_{23}=45^\circ$ , $\theta_{13}=0$                                                                |
| v10_projection_involutions.py      | $Q, \Sigma$ algebra: $K=R+Q\Sigma$ , $L=R+Q(I+\Sigma)$ ; $\chi_Q, \chi_K$ ; det ladder $3\cdot 4\cdot 8\cdot 20=1920= W(D_5) $ ; $a^\top K a=41$                                                                                         |
| v11_unique_KQ.py                   | $K, Q$ are the <i>unique</i> nonneg-int matrices with their compiler budgets, spectrum and monotone rows (enumeration)                                                                                                                   |
| v12_mass_generation_polynomials.py | sector/generation polynomials of $K$ and the anchor-block det ladder $(9, 10, 16, 40)$                                                                                                                                                   |
| v13_open_gates.py                  | gate closures: $M=41=10b_1=\sum L+N_\Phi$ ; $Q_+=A_3$ exponents, $Q_-^2=N_{\text{fam}}$                                                                                                                                                  |
| v14_carrier_uniqueness.py          | $g_{\text{car}}=5$ unique ( $N_{\text{fam}}\in\mathbb{Z}^+$ , $\text{rank } E_8=8$ ); split $(3, 2)$ from $b+s=5$ , $b^2+s^2=13$ ; $\text{Tr } Y=\text{Tr } Y^3=0$                                                                       |
| v15_bootstrap_classification.py    | $D_5\oplus A_3$ unique familyful cyclic-glue of $E_8$ (16-spinor + 3 families), glue $\mathbb{Z}_4$                                                                                                                                      |
| v16_solar_dual_anchor.py           | $a^\top R^{-1}=a^\top L^{-1}=(-\frac{1}{2}, -\frac{1}{2}, 1)$ ; $\sin^2 \theta_{12}=\frac{1}{3}-\frac{\varphi_0^{\text{ret}}}{2}$ ; full PMNS                                                                                            |
| v17_hexagonal_resolvent.py         | finite resolvent $D_y^{-1}=(\sum y^{5-m}\delta^m U_6^m)/(y^6-\delta^6)$ (quark $c$ backbone)                                                                                                                                             |
| v18_quark_yukawa.py                | quark source ratios $\frac{m_u}{m_d}=0.470085$ , $\frac{m_c}{m_s}=13.61$ , $\frac{m_t}{m_b}=40.80$ exact; lepton $c=(\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$ ; full hierarchy                                                           |
| v19_monodromy_moduli.py            | exact pole $z_\star=\frac{794-7\sqrt{9961}}{2187}$ ( $3^7$ ); $D_4$ $SU(3)$ monodromy constructible but $D_4$ -fixed locus positive-dim $\Rightarrow$ exact quark $c$ reduce to $(U)$                                                    |
| v20_lepton_c_derivation.py         | lepton $c$ 's derived: $\delta=\frac{1}{2}$ resolvent $c= \mu_4 ^w/(\frac{5}{4}-\cos\frac{r\pi}{3})\Rightarrow\frac{16}{7}, \frac{4}{3}$ ; product $\frac{2^{g_{\text{car}}}}{N_{\text{fam}}^2}=\frac{32}{9}\Rightarrow\frac{7}{6}$      |
| v21_solar_product_quark.py         | solar coeff $=q(A_3)=\frac{3}{4}$ (glue-norm, $q(D_5)+q(A_3)=2$ ), $\sin^2 \theta_{12}=\frac{1}{3}-\frac{\varphi_0^{\text{ret}}}{2}$ ; product $\frac{2^{g_{\text{car}}}}{N_{\text{fam}}^2}$ ; quark law fails ( $\frac{1}{7}$ vs 0.724) |
| v22_open_gates_audit.py            | residual-gates audit contract: exact reduction of each open gate (A2,B3,B4,B5,B6,C7) machine-pinned (forced part vs. named residual)                                                                                                     |
| v23_anchor_generator.py            | anchor $a=(1, 1, 2)$ : $e_k=(4, 5, 2)$ , $c_3=\frac{1}{2e_1\pi}$ ; $p_n=2+2^n\Rightarrow 240, 248$ , binary ladder; inputs $\rightarrow \{a, \pi\}$                                                                                      |
| v24_quark_ratio_closure.py         | quark ratios $\frac{55}{117}, \frac{34}{47}, \frac{3}{26}$ from TFPT blocks; mass ratios match $< 0.03\%$ ; rational $c$ gauge, $\Lambda_q$ in $O(1)$                                                                                    |
| v25_frontier_conjectures.py        | conjectures: Koide $Q_{\text{src}}+\frac{\varphi_0^{\text{ret}}}{24}\approx\frac{2}{3}$ ; axion $f_a=M_{\text{scal}}/128\rightarrow m_a\approx 23.8\mu\text{eV}$                                                                         |
| v26_flavor_frontier_unification.py | the “11” is not uniquely forced ( $\Rightarrow$ [P]); flavor frontier unifies to one $(U)$ gate; cusp config on the parabolic stability wall                                                                                             |
| v27_wall_representative.py         | explicit balanced wall rep $W_{\text{wall}}$ (cols = $\text{perm}(0, 1, 2)$ , rows $w=(2, 1, 1)$ , $\text{pardeg } 0$ ); selector $\det R=8$ , $\text{Spec}(Q_+)=\{1, 2, 3\}$                                                            |
| v28_gravity_fR.py                  | $R+R^2$ closed: $f(R)$ , $M=c_3^{7/2}\bar{M}_{\text{Pl}}$ scalaron, $N_\star=57\rightarrow n_s, r, A_s$ ; full metric measure open; $248c_3^2<6\log\frac{3}{2}$                                                                          |
| v29_research_contract_             | research-contract certificates: $C_U^{(1)}$ wall enumeration ( $1296\rightarrow 144\rightarrow 5$ orbits; symmetry alone does not pick $W_{\text{wall}}$ ) and $G5$ gap-dominance $2\ V\ <\Delta$                                        |



**How to run.** From the repository root, `python verification/run_all.py` (dependencies `mpmath`, `numpy`, `sympy`); each module also runs standalone, e.g. `python verification/v1_e8_glue.py`. The runner exits with code 0 iff all checks pass; see `verification/README.md` for the full claim $\leftrightarrow$ script map and `verification/status_ledger.csv` for the **single status ledger** (claim ID, status, location, dependencies, script, external datum — the source of truth all six documents are written against). The [F] (Lean-formalised) carrier material lives in the canonical shipped folder `lean4-carrier-rigidity/` (at the package root; the source repository keeps the same project under `experiments/`).